
Oleart y Chen

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Proves inicials

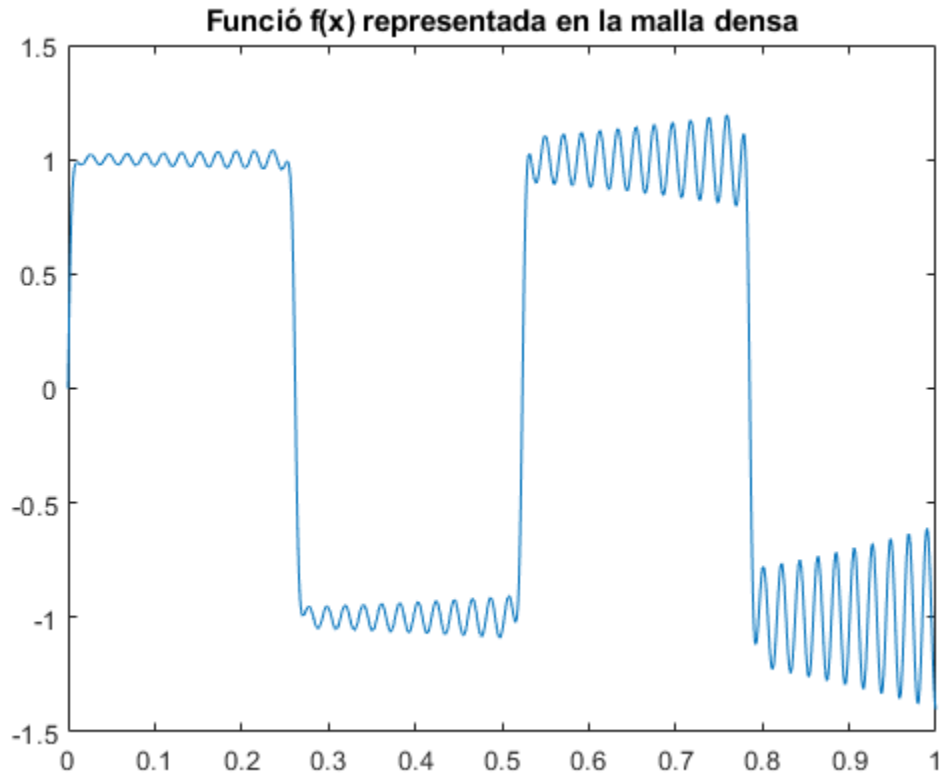
```
% Nodes de Chebychev
n = 100;
c = cheby(n);
nod_chev = 0.5*(c+1); % a = 0, b = 1
nod_equi = linspace(0, 1, n+1);

%Malla densa
m=2000;
malla = linspace(10^-12, 1-10^-11, m+1);
```

Apartat a

Plot de la funció sobre la malla

```
plot(malla, f(malla))
title('Funció f(x) representada en la malla densa')
```



Apartat b

```
% Test de la interpolació baricèntrica amb nodes de Chebychev
```

```
n = 1000;
```

```
c = cheby(n);
```

```
nod_chev = 0.5*(c+1); % a = 0, b = 1
```

```
interpoled = interpol_bar(n, nod_chev, malla);
```

```
figure(1)
```

```
plot(malla, interpoled)
```

```
title('Interpolació amb 1000 nodes')
```

```
% Error obtingut
```

```
error = abs(interpoled - f(malla));
```

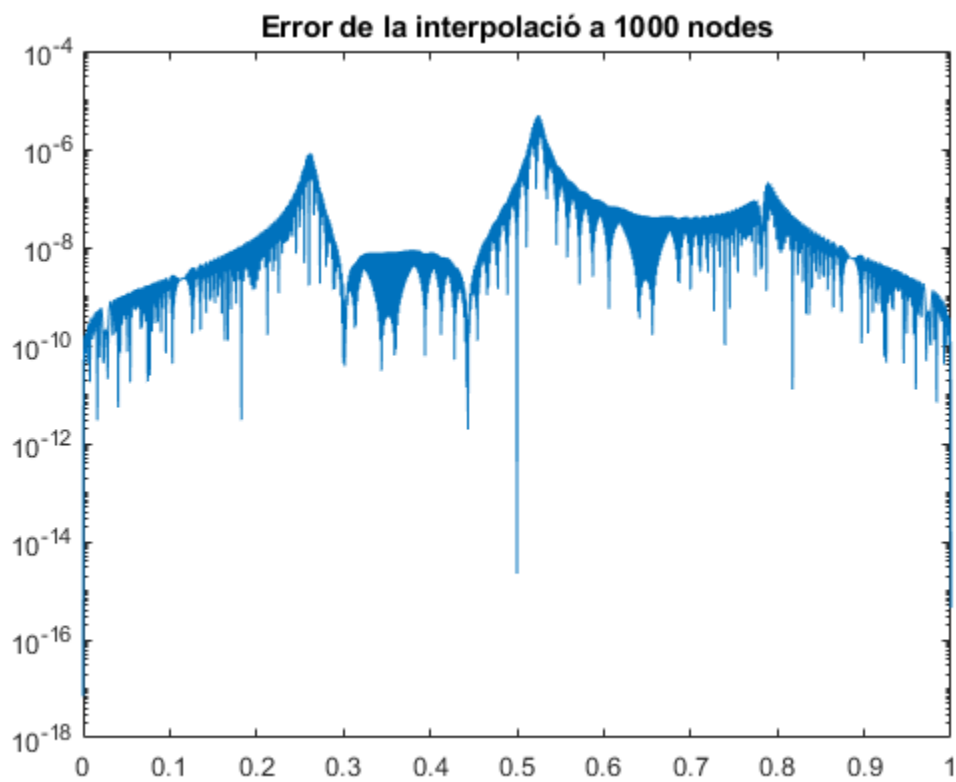
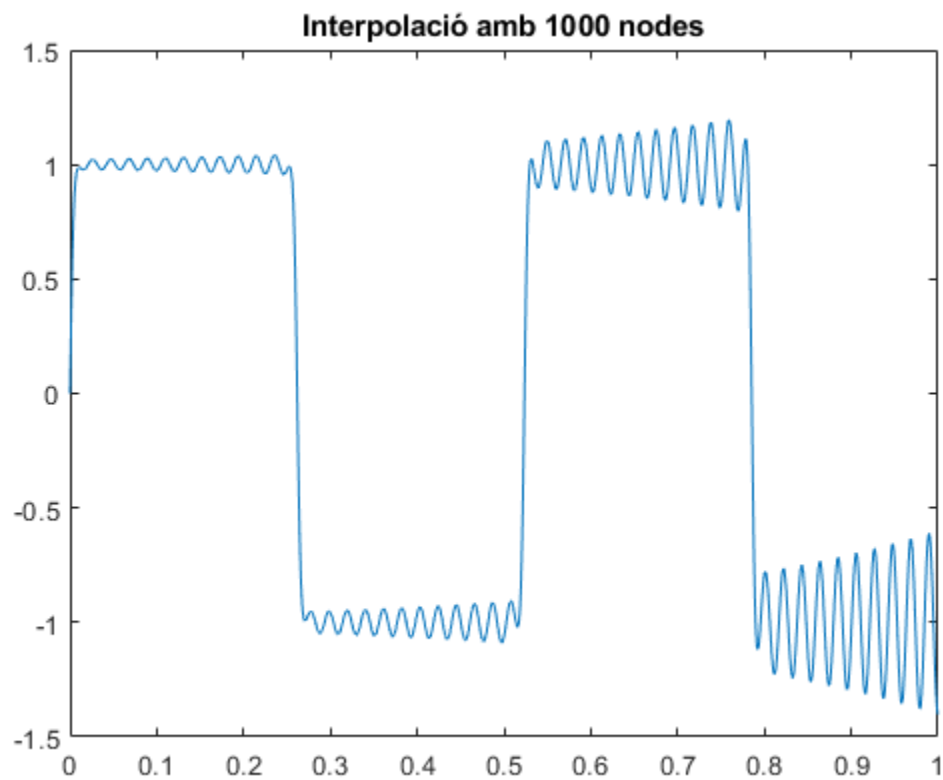
```
disp(max(error))
```

```
figure(2)
```

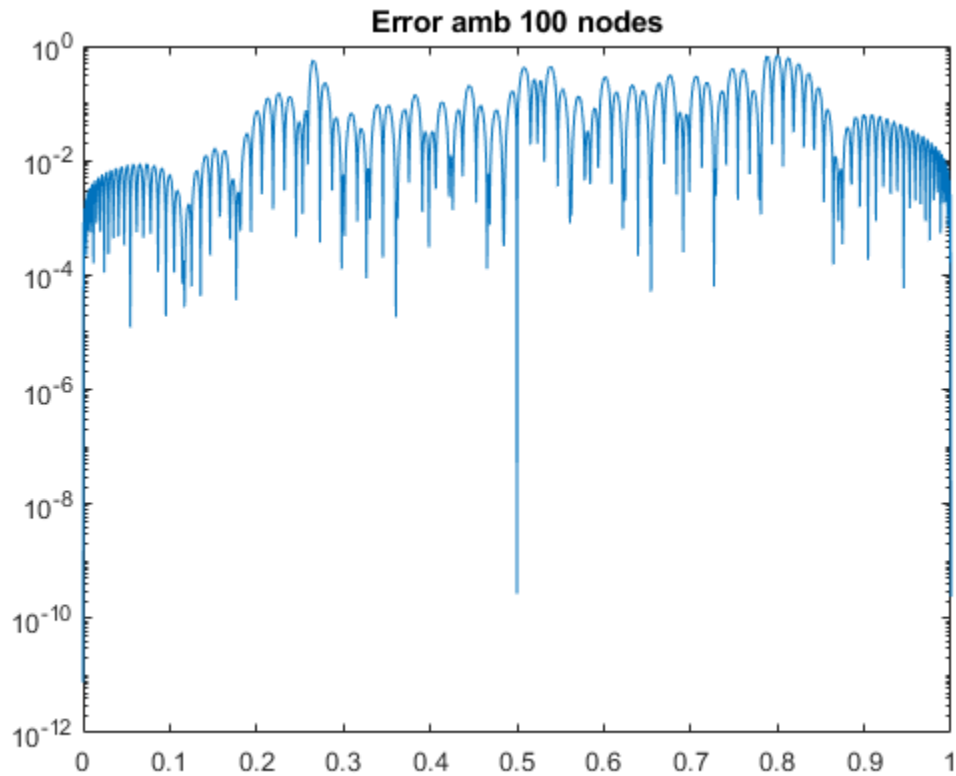
```
semilogy(malla, error)
```

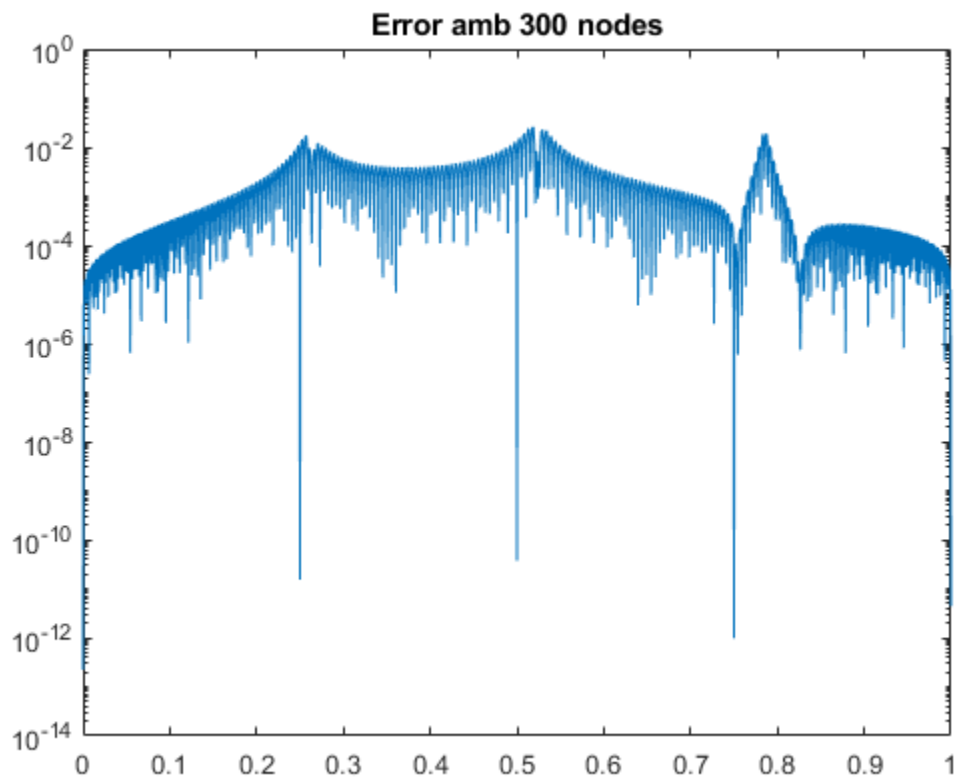
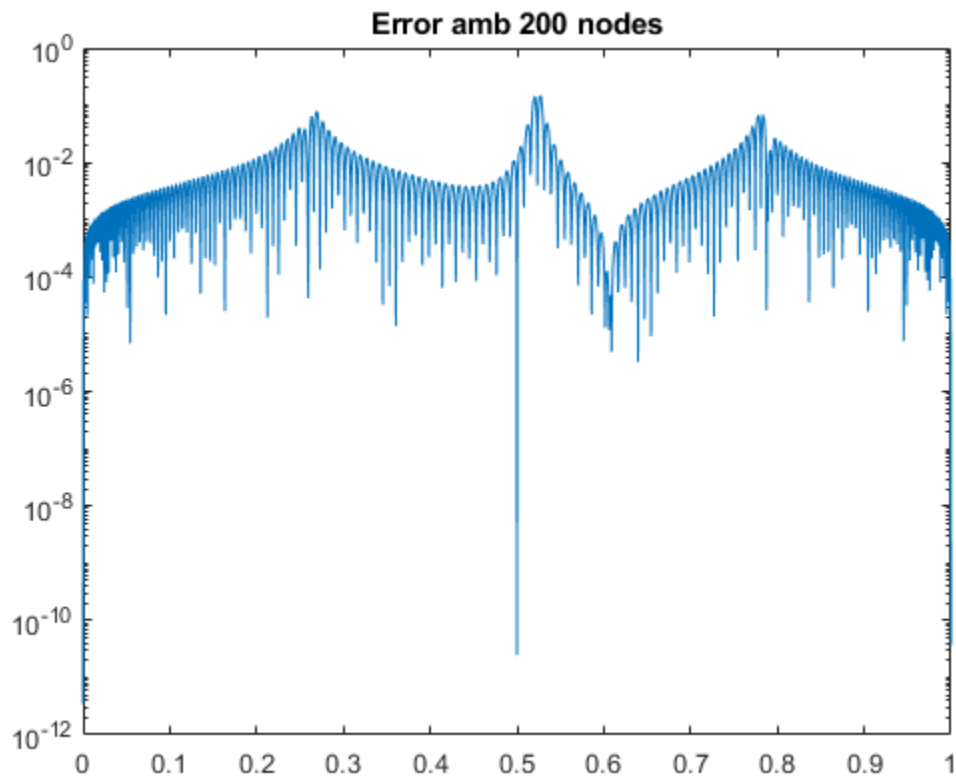
```
title('Error de la interpolació a 1000 nodes')
```

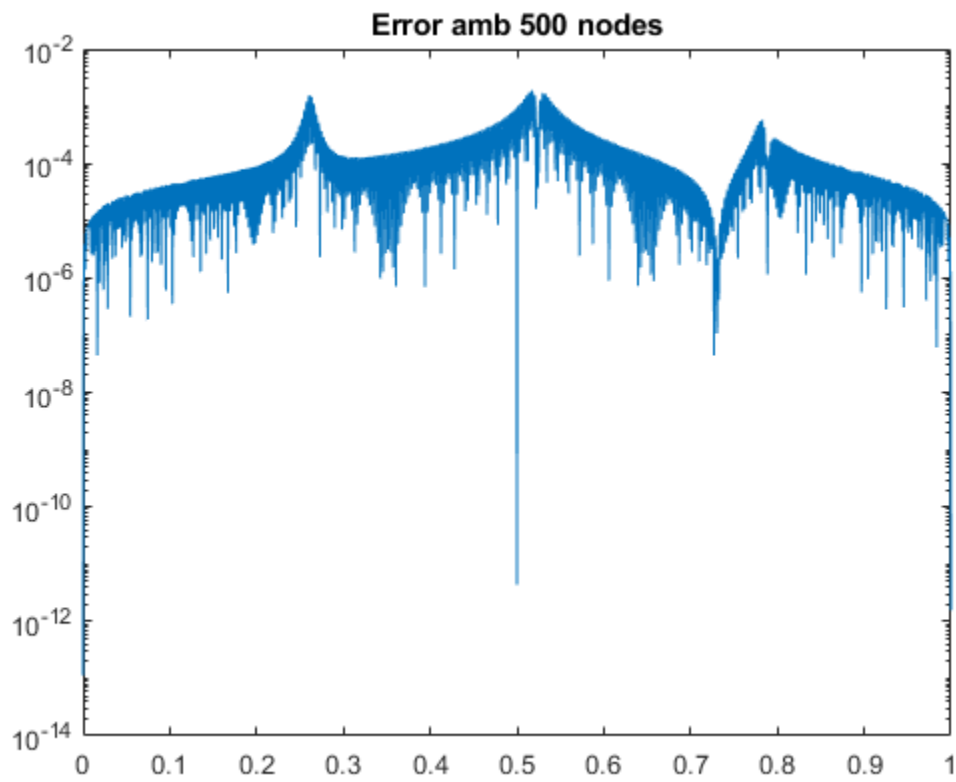
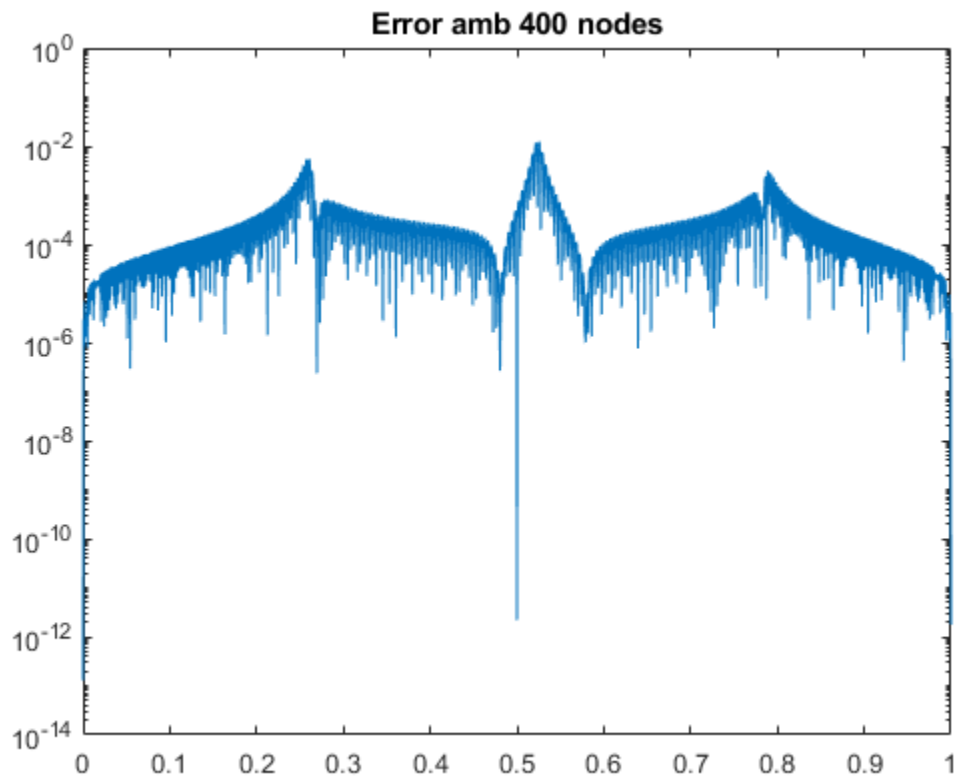
```
4.9230e-06
```



```
%Evolució de l'error en funció del nombre de nodes
for n = 100:100:500
    c = cheby(n);
    nod_chev = 0.5*(c+1); % a = 0, b = 1
    interpolated = interpol_bar(n, nod_chev, malla);
    error = abs(interpoled - f(malla));
    figure(n/100)
    semilogy(malla, error);
    title(sprintf('Error amb %d nodes', n));
end
```







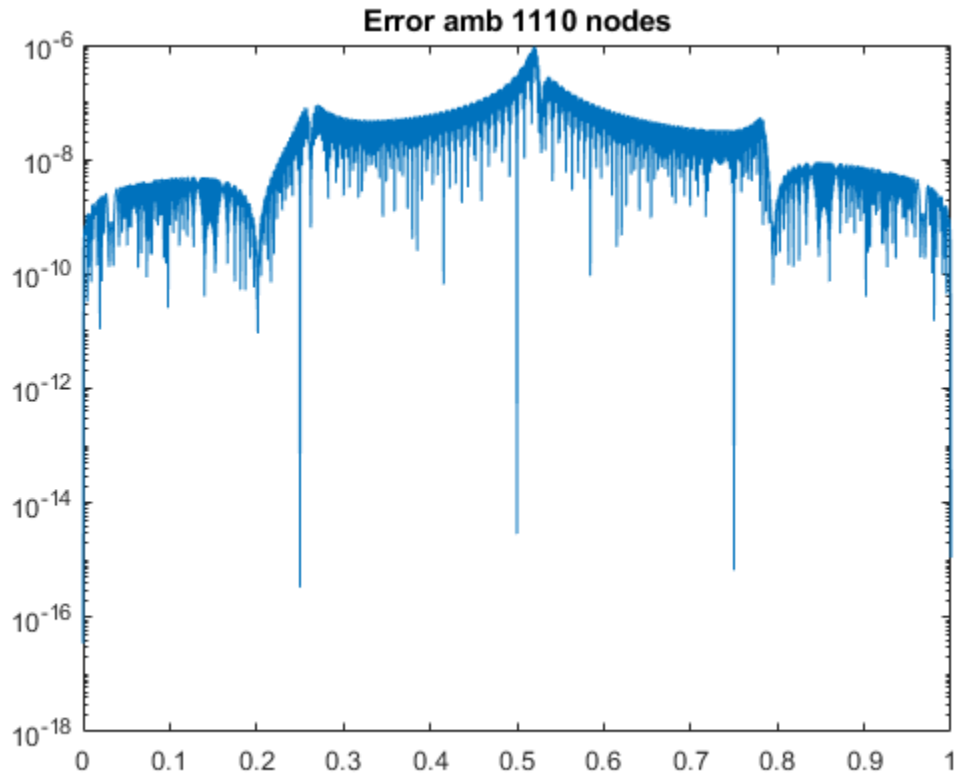
Apartat c

```
error = 100000; %Simulació de l'infinít
n = 0;
while error > 10^-6
    c = cheby(n);
    nod_chev = 0.5*(c+1); % a = 0, b = 1
    interpolated = interpol_bar(n, nod_chev, malla);
    lst_error = abs(interpoled - f(malla));
    if max(lst_error) < error
        error = max(lst_error);
    end
    n = n + 10; % Increments de 10 per fer el codi més hagil
end
disp(['Nombre de iteracions: ', num2str(n)])

% Plot del graf per aquesta n
c = cheby(n);
nod_chev = 0.5*(c+1); % a = 0, b = 1

interpoled = interpol_bar(n, nod_chev, malla);
error = abs(interpoled - f(malla));
disp(['Error màxim: ', num2str(max(error))])
figure(2)
semilogy(malla, error)
title(sprintf('Error amb %d nodes', n));

Nombre de iteracions: 1110
Error màxim: 9.5479e-07
```



Funcions

```
%{
% n: numallaero de nodos
% node: conjunto de nodos en una cierta distribución
% malla: 1 punto de la mallaalla densa
function g = interpol_bar(n, node, m)
    len_m = size(m, 2);

    s = (-1).^(0:n);
    s_n = s.*[f(node(1))/2, f(node(2:end-1)), f(node(end))/2];
    s_d = [s(1)/2, s(2:end-1), s(end)/2];
    g = [];
    for k= 1: len_m
        num = s_n*((m(k)-node).^(-1))';
        den = s_d*((m(k)-node).^(-1))';
        g = [g, num/den];
    end
end

function f = f(x)
    f = tanh(20*sin(12*x))+(2/100)*exp(3*x).*sin(300*x);
end
```

```
function c = cheby(n)
    c = [];
    for j = 0:n
        c = [c, cos((j*pi)/n)];
    end

    %}
```

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